

The constant regions in the heavy chain of antibodies dictate the molecule's

Epitope

0%

Fab fragment

0%

Variable domain

0%

Isotype

0%

Which of the following Abs crosses the placenta and provide passive immunity to the fetus?

IgA

0%

IgM

0%

IgE

0%

IgG

0%

The epitope-binding site on the antibody is located ____

on the Fab fragment

0%

on the Fc fragment

0%

on the constant region

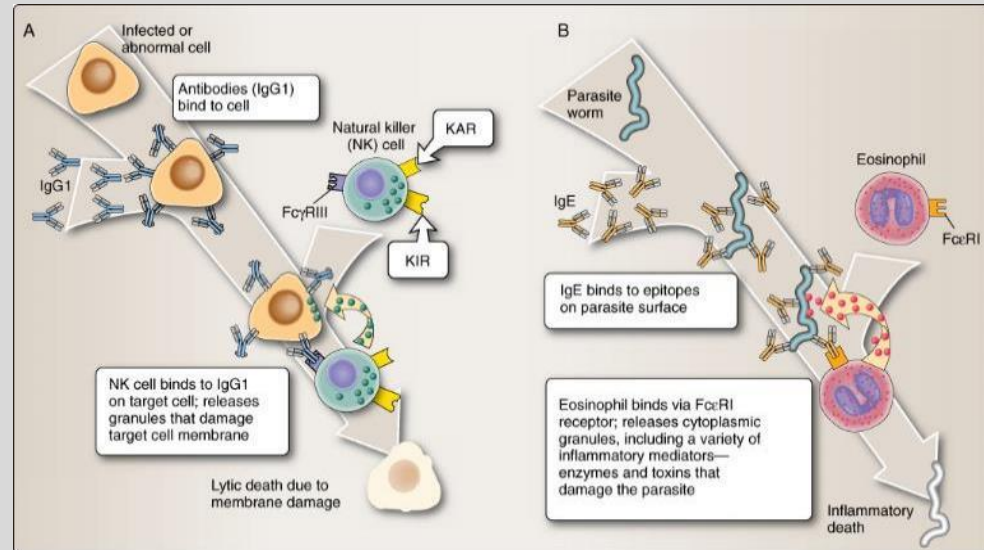
0%

on the variable region

0%

Ag-Ab reactions

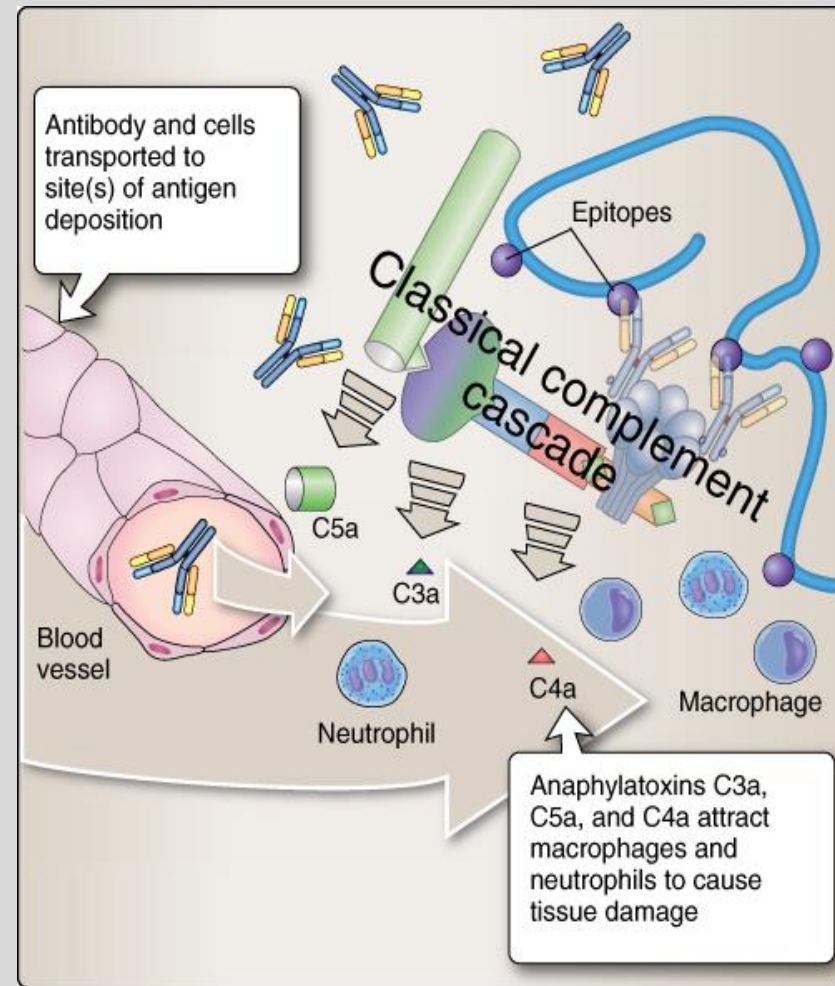
- Antibody-dependent cell-mediated cytotoxicity:
 - Ab-tagged bacteria, parasites or worms attract NK cells & eosinophils



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Hypersensitivities (Ab-mediated)

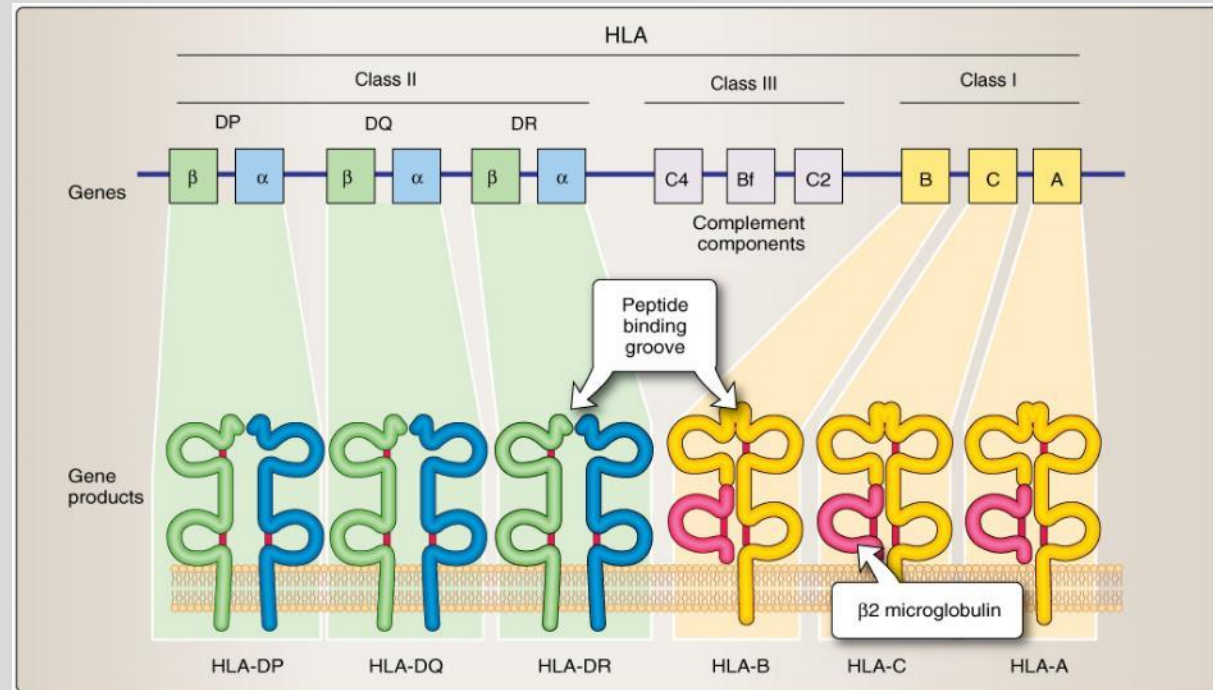
- Type III hypersensitivity (immune complex):
 - Circulating Ag-Ab complexes (IgG) → vasculitis
 - Localized (Arthus): abscess or necrotizing vasculitis
 - Systemic (serum sickness):
 - a) Exogenous: non-human Abs/drug induced (e.g. penicillin)
 - b) Endogenous: e.g. SLE



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

MHC molecules

- MHC=HLA
- 3 classes:
 - a) MHC I
 - b) MHC II, &
 - c) MHC III
(*complement components*)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

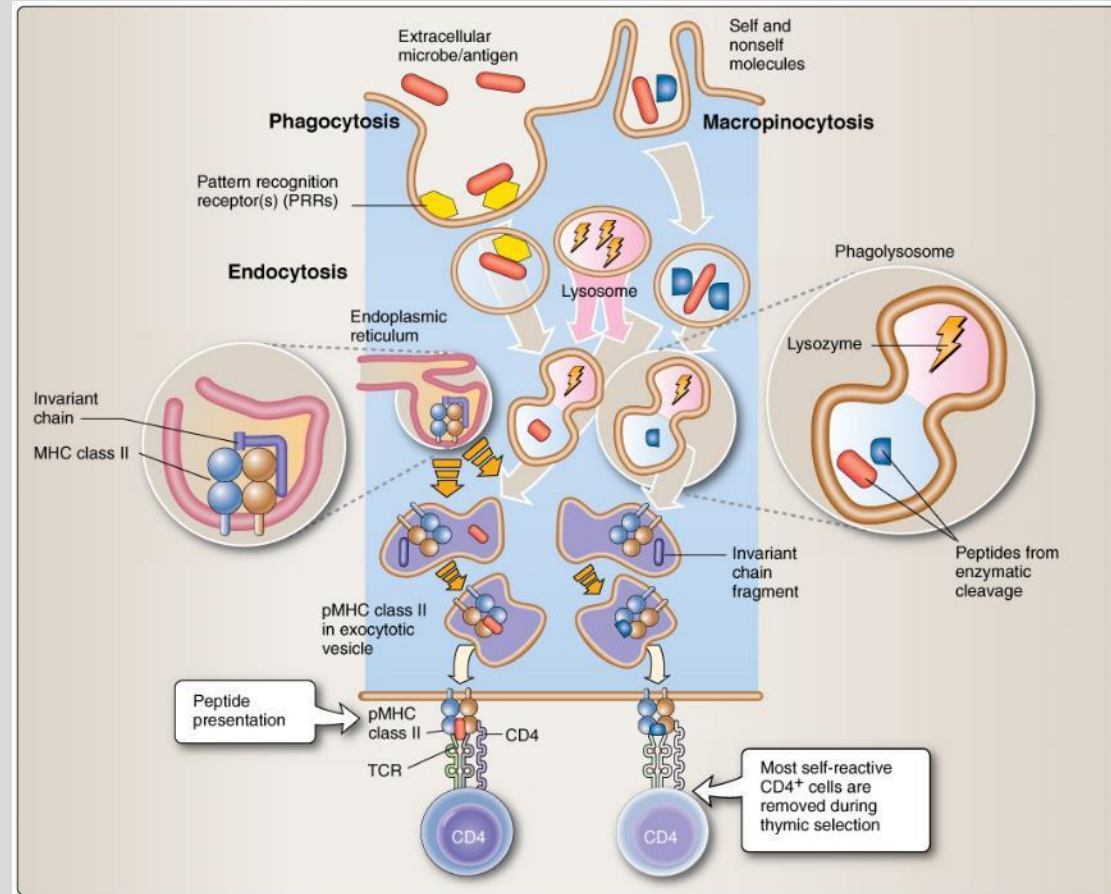
MHC molecules

- MHC I:
 - a) **All nucleated cells**
 - b) Function: self reporting (viral-infected & tumors)
 - c) HLA-A, B & C (100 alleles at each locus)

- MHC II:
 - a) Expressed mostly on APCs (DC, Macrophages)
 - b) But also B-cells; some activated T-cells, some specialized epithelial cells in the thymus & intestine)
 - c) Function: involved in antigen presentation

MHC II presentation

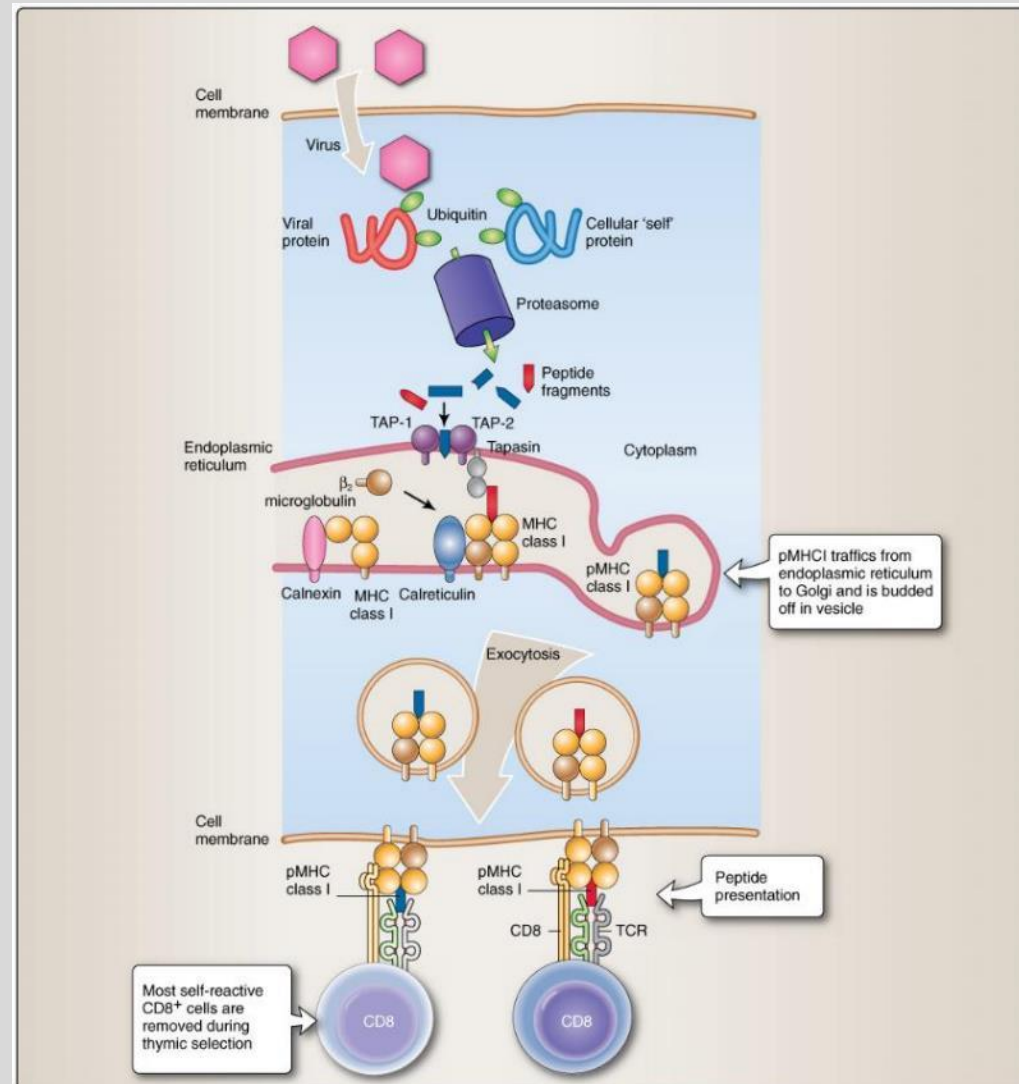
- DC main APC (sentinels)
- Actin-dependent phagocytosis
- PRRs engage PAMPs (extracellular Ags)
- Ab or C-tagged Ags
- Ag displayed on MHC II
- MHC II displays self & non-self Ag (tolerance)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

MHC I presentation

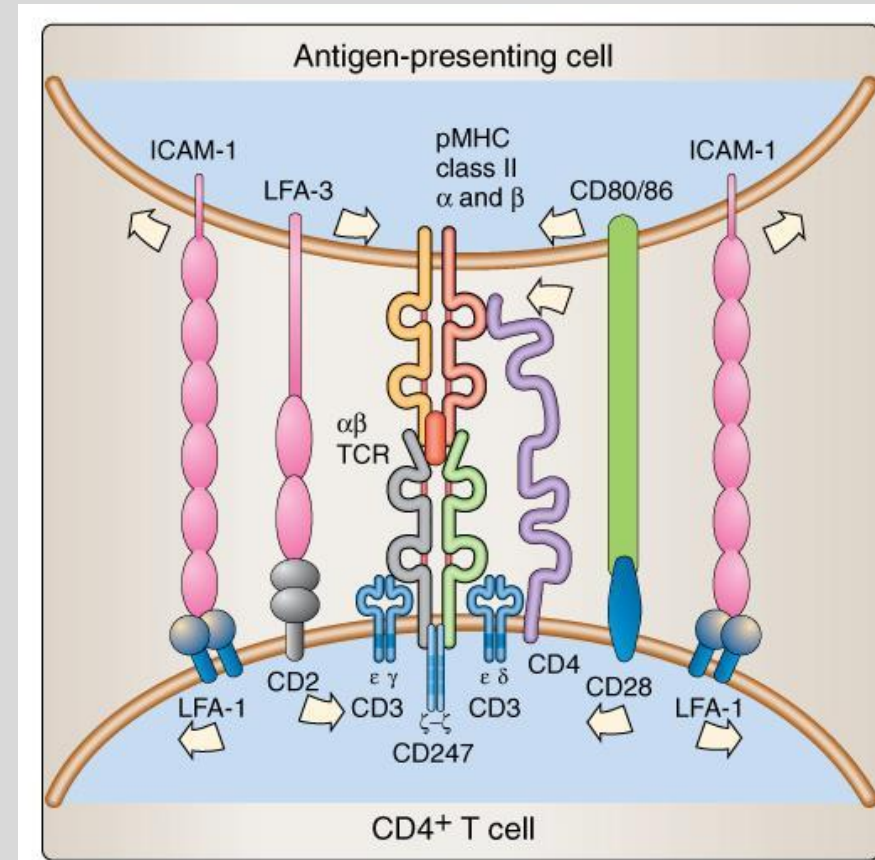
- Intracellular Ags tagged with ubiquitin → proteasome destruction → ER → Golgi → display on MHC I → recognition by CD8+ cells



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

T-helper cell activation (CD4⁺)

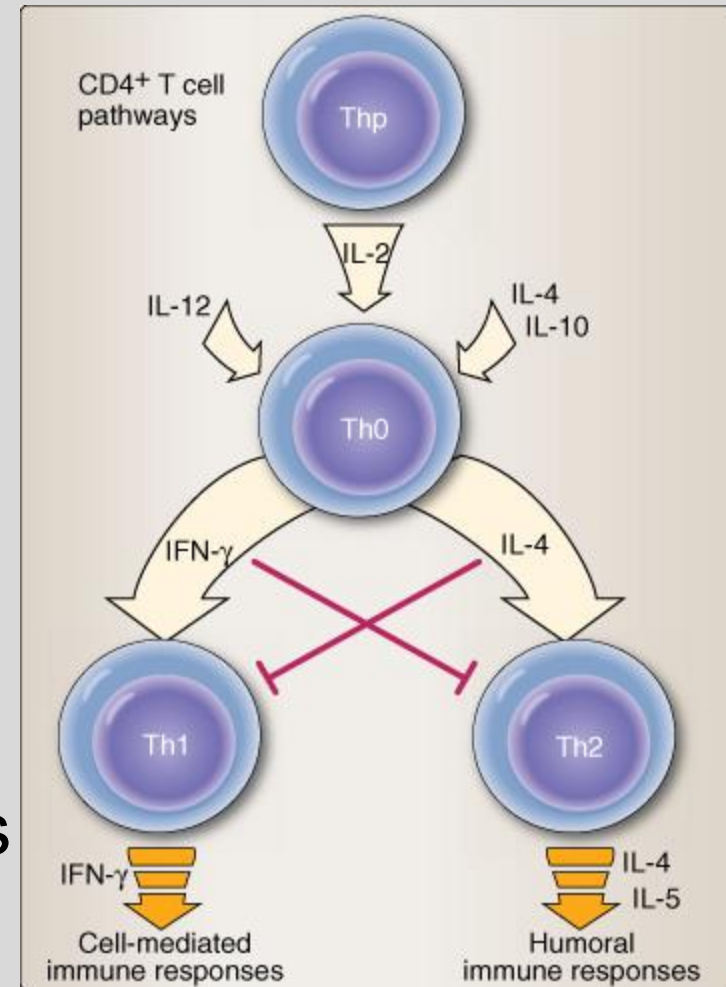
- Only recognize MHC II-associated Ags
- Immunologic synapse: interface between naïve Th & APC
- **Stabilized by CD28** (*costimulatory*) → signal transduction
- Absence of costimulatory signal = anergy/apoptosis



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

T-helper cell activation (CD4+)

- 1st contact with Ag → prime T-helper
- T-helper → Th1, Th2 or Th17 (*cytokines from APC*)
 1. Th1: IFN- γ /IL-12
 2. Th2: IL-4/IL-10
 3. Th17: IL-17/IL-6/TGF- β
- Some become memory T-cells (*high expression of CD28*)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Receptors on T-cells are typically associated with?

CD55

0%

CD3

0%

IL-4

0%

Ubiquitin

0%

B-cell activation

- Recognize Ags in association with MHC class II or without
- Ag binds to BCR → endocytosis → Ag presentation on MHC to CD4+ (T-dependent Ags) → Th2/Th17 (IL-4/IL-6) → plasma cells (some memory)
- T-independent (Ti) Ags (*activate B-cells without T-cells help*)
 - a) Ti-1 Ags: polyclonal activators bind to receptors other than BCR (E.g. LPS); B-cell mitogens
 - b) Ti-2 Ags: multivalent polysaccharides; stimulate only mature B-cells

Memory B-cells undergo point mutations in the coding regions of both light and heavy chains when re-exposed to an antigen. This result in

New Ag specificity

0%

Isotype switching (IgM to IgG)

0%

DNA chromosomal rearrangement to alter specificity

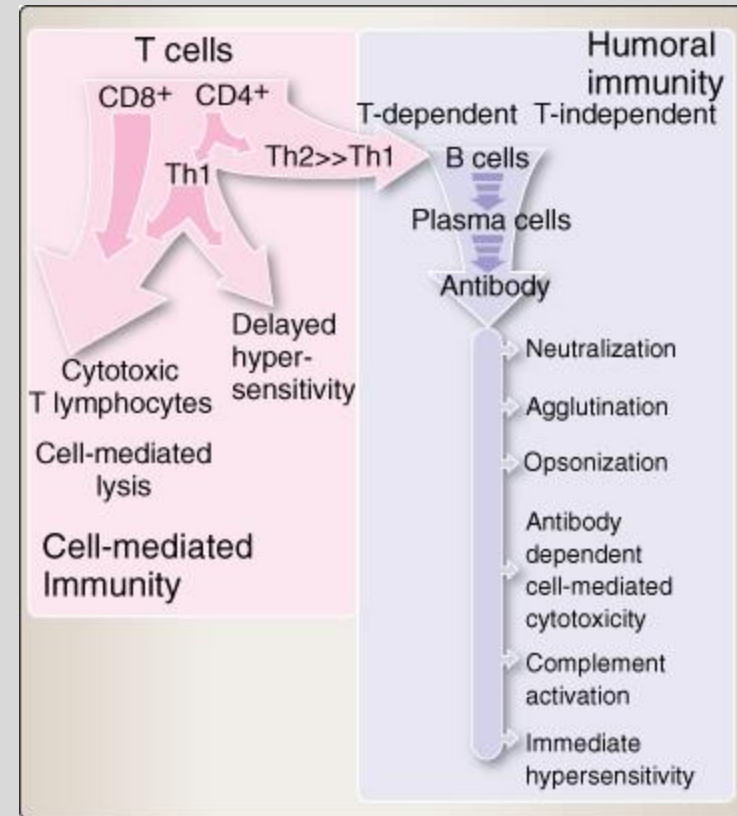
0%

generation of antibodies with increase affinity for its epitope

0%

Two arms of the Adaptive IR

- Humoral IR:
 - a) Abs
 - b) Extracellular pathogens
 - c) Type I, II & III hypersensitivities
- CMI:
 - a) Intracellular pathogens
 - b) Type IV hypersensitivities



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Which of the following is initiated by the interaction of host cell membranes and IgM/IgG but never IgE?

Arthus reaction

0%

Type I hypersensitivity

0%

Type II hypersensitivity

0%

Type III hypersensitivity

0%

The tuberculin skin test is an example of?

Type I hypersensitivity

0%

Type II hypersensitivity

0%

Type III hypersensitivity

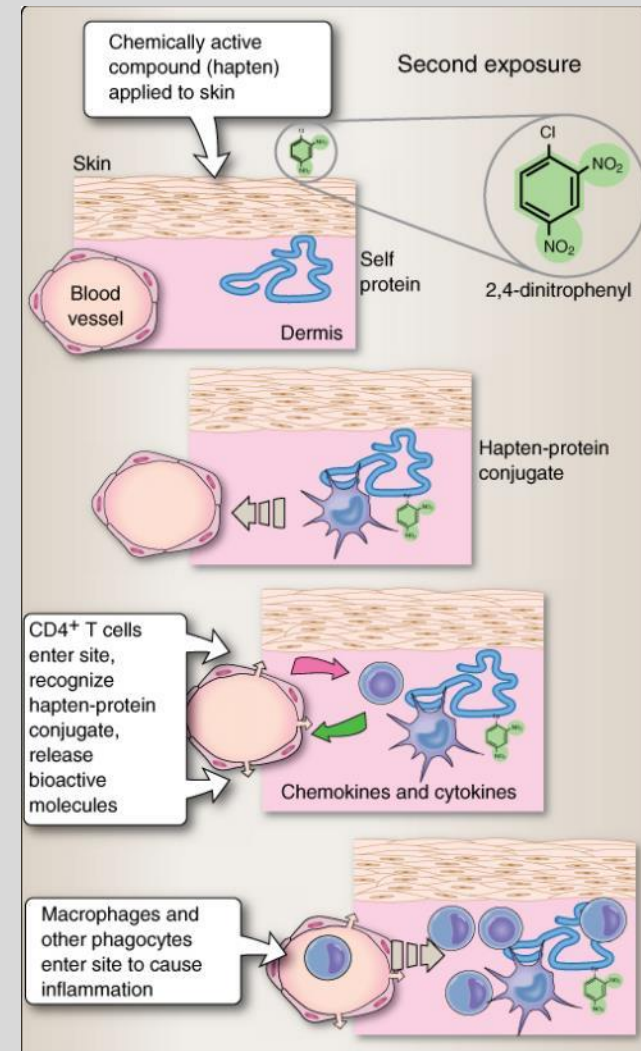
0%

Type IV hypersensitivity

0%

Type IV hypersensitivity

- Th1/Macrophages induced inflammation
 - Topical: contact dermatitis (poison Ivy, TNT, nickel)
 - Ag injected or DTH (48 to 72 hours): due to microbes (*Mycobacteria*, *Leishmania*, *Candida*, mumps virus, many fungal skin tests)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Adaptive responses

Adaptive Responses Involved in Clearance of Active Infections

Organisms	IgM, IgG, IgA				IgE	CTL	DTH
	Complement Activation	Opsonization	ADCC	Neutralizing Antibody			
Viruses							
Bacteria (intracellular)							
Bacteria (extracellular)							
Protozoa (intracellular)							
Protozoa (extracellular)							
Fungi							
Flatworms							
Roundworms							

Which of the following cells is implicated on the prevention of autoimmune responses?

APC

0%

Anergic T-cells

0%

DC

0%

Treg

0%

Case study

A 4-year-old female was brought to the ED with dyspnea and uncontrolled muscle spasms. On physical examination the patient had difficulty opening her mouth as well as swallowing. History indicates that the child has *not* received standard childhood vaccination. The patient was recently stung by a bee while playing in the school yard. The physician orders laboratory tests that include microscopic analysis as well as anaerobic culture. Microscopic analysis of the Gram stain is shown below.

